SECTION–I(i) : (Maximum Marks : 30)

- This section contains **TEN** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : –1 In all other cases

1. Which of the following set of four quantum numbers is possible?

(A) n = 4, ℓ = 3, m = 2, s = $+\frac{1}{2}$ (B) n = 3, ℓ = 3, m = 0, s = $+\frac{1}{2}$

(C) n = 2, ℓ = 1, m = –2, s = $+\frac{1}{2}$ (D) n = 5, ℓ = 3, m = +4, s = $+\frac{1}{2}$

2. Which of the following order for given property is **WRONG** –
(A) $O^{2-} > O > O^{+}$ (size)
(B) Mg < S < Mn (valence electron)
(C) $Fe^{2+} < Mn^{2+} < Cr^{3+}$ (total number of unpaired electron)
(D) Sc < Y < La (atomic size)

3. The order of which of the following oxides is arranged according to decreasing basic nature ?
(A) Na_2O, MgO, Al_2O_3 (B) Al_2O_3, MgO, Na_2O
(C) Al_2O_3, MgO, Na_2O (D) MgO, Na_2O, Al_2O_3

4. Select the **CORRECT** radii order –
(A) K > Ca > P > Mg > N (B) $Cl^-_{(g)} > S^{2-}_{(g)} > F^-_{(g)} > Na^+_{(g)}$
(C) $Be^{2+}_{(aq)} > Li^+_{(aq)}$ (D) Co ≃ Ni ≃ Zn

5. Select the **CORRECT** ionisation energy order.
(A) Ag < Cu < Au (B) O < N < F (C) La < Y < Sc (D) All of these

6. Which of the following conversion is exothermic?
(A) $Na \rightarrow Na^+$ (B) $Li \rightarrow Li^-$ (C) $N \rightarrow N^-$ (D) $Mg \rightarrow Mg^-$

7. The first seven successive ionisation energy (in eV/atom) of element 'X' are 8 eV/atom, 16 eV/atom, 33 eV/atom, 45 eV/atom, 166 eV/atom, 205 eV/atom and 246 eV/atom respectively. The element 'X' can be
(A) C (B) O (C) N (D) Si

8. Lithium acts as best reducing agent in aqueous medium because
Reason-1 : Lithium has high sublimation energy.
Reason-2 : Lithium has large I.E. among group 1 metals.
Reason-3 : Li^+ ion has highest hydration energy among group 1 metal ions.
Which of the above reason is correct.
(A) Reason 1 alone (B) Reason 1 and 3 both
(C) Reason 3 alone (D) No reason is correct

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9. Which of the following order is **INCORRECT**.

(A) $C\ell^- < K^+ < Ca^{+2}$ (I.E.)

(B) Pd < Ag (I. E.)

(C) $Li^+ > Na^+$ (E.A.)

(D) Mg^{+2} (aq) < Ca^{+2} (aq) (ionic mobility)

10. Which of the following is **NOT** a result of d-contraction or f-contraction?

(A) Greater I.E. of Ga compared to Aℓ.

(B) Smaller size of Ga compared to Aℓ.

(C) Hg has highest I.E. among : Zn, Cd, Hg

(D) Exceptional configuration of Lanthanum.

SECTION-I(ii) : (Maximum Marks: 20)

- This section contains **FIVE** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : –2 In all other cases.

11. Which of the following is/are the **INCORRECT** match for atom of elements?

(A) [Rn] $6d^2\ 7s^2 \rightarrow$ d-block, 4th group

(B) [Xe] $4f^{14}\ 5d^1\ 6s^2 \rightarrow$ d-block, 3rd group

(C) [Xe] $4f^7\ 6s^2 \rightarrow$ f-block, 9th group

(D) [Kr] $4d^{10} \rightarrow$ d-block, 12th group

12. Choose the **CORRECT** statement(s) :

(A) Smallest anion in the periodic table in H^-

(B) Ionic mobility of $Mg^{2+}_{(aq)}$ is greater than that of $Be^{2+}_{(aq)}$

(C) Electronegativity of Pb^{2+} is more than that of Pb^{4+}

(D) Second ionisation of Al is higher than second ionisation of Mg.

13. The **CORRECT** order of electron affinity of the element or ion is/are?

(A) O > Se (B) $F^+ > Cl^+$ (C) P > N (D) Ne > Ar

14. Which of the following oxide(s) will react with H_2SO_4 as well as KOH solution to produce salt?

(A) Al_2O_3 (B) BaO (C) SnO (D) ZnO

15. If an element 'X' belongs from 5th period and 10th group then select the **CORRECT** statement regarding element X.

(A) Last electron of element 'X' enters in n = 4 and ℓ = 2

(B) It is a diamagnetic specie

(C) It is not a transition element

(D) It is a coinage metal

SECTION–I(iii) : (Maximum Marks : 03)

- This section contains **ONE** question.
- **Each question has matching lists.** The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : –1 In all other cases

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16. Match list-I with list-II

List-I (Periodic properties)	**List-II (Order)**
(P) Ionisation energy	(1) Sn < Pb
(Q) Radii	(2) N < C < O
(R) Electronegativity	(3) Li < Be < B
(S) Electron affinity	(4) $O^- < S^-$

Select the correct code :

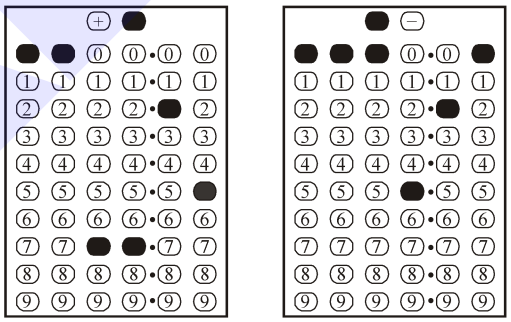
Code :

	P	**Q**	**R**	**S**
(A)	4	1	3	2
(B)	1	2	3	4
(C)	4	3	2	1
(D)	1	4	2	3

SECTION-II : (Maximum Marks: 09)

- This section contains **THREE** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

For Example : If answer is –77.25, 5.2 then fill the bubbles as follows.



- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

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1. From the following list of enthalpies, find the number of enthalpies which are **NOT** required to prepare Born-Haber cycle of NaI

ΔH_{Sub} (Na), ΔH_{Sub} (I_2), ΔH_{Diss} (I_2), ΔH_{Hyd} (Na^+),
ΔH_{Hyd} (I^-), $\Delta H_{I.E.}$ (I), ΔH_{eg1} (I), $\Delta H_{L.E.}$ (NaI),
$\Delta H_{I.E.}$ (Na^+)

2. In how many following pairs, first species has higher ionisation energy than second species :

(i) Na^+, Mg^{2+} (ii) S, Cl (iii) Cu, Zn (iv) Xe, Kr

(v) B, Be (vi) O^{2-}, O (vii) Al, Si (viii) Cl^-, Cl

3. In aqueous solution, how many following species have higher hydration energy than Li^+.

Na^+, K^+, Cl^-, F^-, Rb^{+1}, Be^{2+}

SECTION–IV : (Maximum Marks : 16)

- This section contains **TWO** questions.
- Each question contains two columns, **Column-I** and **Column-II.**
- **Column-I** has **four** entries (A), (B), (C) and (D)
- **Column-II** has **five** entries (P), (Q), (R), (S) and (T)
- Match the entries in **Column-I** with the entries in **column-II.**
- One or more entries in **Column-I** may match with one or more entries in **Column-II.**
- The ORS contains a 4 × 5 matrix whose layout will be similar to the one shown below :

(A) (P) (Q) (R) (S) (T)

(B) (P) (Q) (R) (S) (T)

(C) (P) (Q) (R) (S) (T)

(D) (P) (Q) (R) (S) (T)

- For each entry in **column-I,** darken the bubbles of all the matching entries. For example, if entry (A) in **Column-I** matches with entries (Q), (R) and (T), then darken these three bubbles in the ORS. Similarly, for entries (B), (C) and (D).
- For each question, marks will be awarded in <u>one of the following categories</u> :

For each entry in **Column-I**

Full Marks : +2 If only the bubble(s) corresponding to all the correct match(es) is (are) darkened

Zero Marks : 0 In all other cases

1.

Column-I		**Column-II**	
(A)	$\Delta H < O$ (Exothermic)	(P)	$N^+(g) \rightarrow N^{+2}(g)$
(B)	$\Delta H > O$ (Endothermic)	(Q)	$N(g) \rightarrow N^-(g)$
(C)	$\Delta H_{IE_2}(N)$	(R)	$N(g) \rightarrow N^{+2}(g)$
(D)	$\Delta H_{Eg_1}(N)$	(S)	$N^-(g) \rightarrow N^{-2}(g)$
		(T)	$F^-(g) \rightarrow F^-(aq)$

2.

Column-I (Ionic Compound)		**Column-II (Properties)**	
(A)	Li_2O	(P)	Highest lattice energy among given compounds
(B)	Na_2O	(Q)	Basic oxide
(C)	MgO	(R)	Amphoteric oxide
(D)	$A\ell_2O_3$	(S)	Least lattice energy among given compounds
		(T)	The gaseous cation (among given ionic ompounds) will be most extensively hydrated in aqueous medium